Homework 7

1. a) Calculate the wavelength of an electron with kinetic energy of 1 meV.

$KE = 1E\text{-}3\,eV$

$m_e c^2 = 5E5\,eV$

$\hbar c = 2E\text{-}7\,eV\cdot m$

$$E = \frac{p^2}{2m} = \frac{(\hbar c)^2 k^2}{2m} = \frac{(\hbar c)^2\left(\frac{4\pi^2}{\lambda^2}\right)}{2mc^2} \Rightarrow \frac{E\,2mc^2}{(\hbar c)^2} = \frac{4\pi^2}{\lambda^2} \Rightarrow \frac{1}{\lambda^2} = \frac{E\,2mc^2}{4\pi^2(\hbar c)^2} \Rightarrow \lambda = \sqrt{\frac{4\pi^2(\hbar c)^2}{E\,2mc^2}}$$

$$k = \frac{\sqrt{2mc^2E}}{\hbar c} \Rightarrow \frac{2\pi}{\lambda}$$

$$= \frac{2\pi(\hbar c)}{\sqrt{E\,2mc^2}}$$

$$\hookrightarrow \frac{2\pi(2E\text{-}7)\ eV\cdot m}{\sqrt{1E\text{-}3\,eV\cdot 2(5E5\,eV)}} = 3.974E\text{-}8\,m = 39.7\,nm$$

b) How much does the phase of this electron wave advance if it moves 1 nm?

$$\phi = kx + \phi_0$$

$$\hookrightarrow \Delta\phi = k\Delta x + \phi_0 = \frac{2\pi\Delta x}{\lambda} + 0 \Rightarrow \frac{2\pi(1E\text{-}9\,m)}{39.7E\text{-}9\,m} = 0.158\ rad = 9.1^\circ$$

2. a) An electron is in the 3rd state from the bottom (second "excited" state) in a perfect one-dimensional square well with size 1 nm. What is its kinetic energy?

$$E_n = \frac{N^2\pi^2\hbar^2}{2mL^2} \Rightarrow \frac{N^2\pi^2(\hbar c)^2}{2(mc^2)L^2} \Rightarrow \frac{9\pi^2(2E\text{-}7\,eV\cdot m)^2}{2(5E5\,eV)(1E\text{-}9\,m)^2} = 3.55\,eV$$

b) Suppose now that the electron is in a *three*-dimensional box with sides 1 nm, 2 nm, and 3 nm. What is the energy of its ground state?

$$E_{n_{xyz}} = E_{n_x} + E_{n_y} + E_{n_z} \Rightarrow \frac{N^2\pi^2(\hbar c)^2}{2m}\left(\frac{1}{L_x^2} + \frac{1}{L_y^2} + \frac{1}{L_z^2}\right) = \frac{1\pi^2(2E\text{-}7\,eV\cdot m)^2}{2(5E5\,eV)}\left[\frac{1}{1E\text{-}9\,m^2} + \frac{1}{2E\text{-}9\,m^2} + \frac{1}{3E\text{-}9\,m^2}\right] = 0.511\,eV$$

3. a) Using the formula we derived in class (also given in the "Wave equations" PDF clip on the Canvas site), determine the density of electrons (assuming two equally occupied spin states) if their Fermi energy is 1 eV.

$$E_F = \frac{\hbar^2 k_F^2}{2m} \Rightarrow N = \left(\frac{V}{\pi^2}\right) 2 \cdot \frac{1}{8}\int_0^{k_F} 4\pi k^2\,dk$$

total # of electrons — 2 spin state — one octet of 3D space

$$\hookrightarrow = \left(\frac{V}{\pi^3}\right)\frac{1}{4}\int_0^{E_F} 4\pi \frac{k^{\cancel{2}}m}{\hbar \cancel{k}}\,dE$$

$$= \left(\frac{V}{\pi^3}\right)\frac{1}{4}\int_0^{E_F} 4\pi \frac{\sqrt{2mE}}{\hbar^2}\,m\,dE$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m}$$

$$dE = \frac{\hbar k}{m}dk$$

$$dk = \frac{m}{\hbar^2 k}dE$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

density of particles: $\frac{N}{V} = \frac{2\sqrt{2}}{3\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{3}E_F^{3/2}\right)$, $\frac{N}{V} = n$

$$n = \frac{2\sqrt{2}}{3\pi^2}\frac{(m_e c^2)^{3/2}}{(\hbar c)^3}\left(E_F^{3/2}\right)$$

$$= \frac{2\sqrt{2}}{3\pi^2}\frac{(5E5\,eV)^{3/2}}{(2E\text{-}7\,eV\cdot m)^3}\cdot(1\,eV)^{3/2} = 4.22\times10^{27}\,m^{-3}$$

b) Using the related formula for the total kinetic energy U, determine the average energy per particle. equate num particles to natural energy

$\hbar c = 2\times10^{-7}\,eV\cdot m$

$mc^2 = 5\cdot10^5\,eV$ (for e^-)

$n = 4.22\times10^{27}\,m^{-3}$

$$U = 2V\frac{\sqrt{2}}{2\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{5}E_F^{5/2}\right)$$

$$N = 2V\frac{\sqrt{2}}{2\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{3}E_F^{3/2}\right)$$

$$\text{so, } \frac{U}{N} = \frac{\left[2V\frac{\sqrt{2}}{2\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{5}E_F^{5/2}\right)\right]}{\left[2V\frac{\sqrt{2}}{2\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{3}E_F^{3/2}\right)\right]}$$

$$\hookrightarrow \frac{\left(\frac{2}{5}E_F^{3/2}\right)}{\left(\frac{2}{3}E_F^{5/2}\right)} \Rightarrow \frac{3}{5}E_F = 3/5\,eV = 0.6\,eV$$

c) Suppose that the electrons were given a new "effective mass" equal to half the standard vacuum value. How would that change your answers to (a) and (b)?

a.) $n = \frac{2\sqrt{2}}{3\pi^2}\frac{(m_e c^2)^{3/2}}{(\hbar c)^3}\left(E_F^{3/2}\right) \Rightarrow n^* = \frac{2\sqrt{2}}{3\pi^2}\frac{\frac{1}{2}(m_e c^2)^{3/2}}{(\hbar c^2)}\left(E_F^{3/2}\right)$

$\propto m^{3/2}$ $\qquad \propto \frac{1}{2}(m)^{3/2}$

if $m_e c^* = \frac{1}{2} m_e c^2$ $\quad \Delta n = n - n^* = \frac{n}{2\sqrt{2}} = \frac{4.22 \times 10^{27}\, m^{-3}}{2\sqrt{2}} = 1.492 \times 10^{27}\, m^{-3}$

density decreases by : $4.22 \times 10^{27}\, m^{-3} - 1.492 \times 10^{27}\, m^{-3} = 2.728 \times 10^{27}\, m^{-3}$

b.) since the fermi-energy was given and assumed to be independent of mass, the energy/particle in this volume will stay the same

4. a) In a solid of size 1 cm, what is the difference in the value of k_x from one electron state to the next, assuming that the electron states are nearly plane waves?

grid: $\left(\frac{2\pi}{L_x}N_x, \frac{2\pi}{L_y}N_y, \frac{2\pi}{L_z}N_z\right) \Rightarrow k_x = \frac{2\pi N_x}{L}$ w/ $n_x = 0, \pm 1, \pm 2, \ldots$

- volume in k-space $\quad$ so, $\Delta k_x = \frac{2\pi}{L}$

$= \frac{2\pi}{10^{-2} m} = 200\pi = 6.28 \times 10^2\, m^{-1}$

b) What is the energy difference between two adjacent k-states in this solid?

$E_F = \frac{\hbar^2 k_x^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{2\pi}{L}\right)^2 = \frac{(\hbar c)^2}{2 m_e c^2}\left(\frac{2\pi}{L}\right)^2 \Rightarrow \frac{(2E{-7}\, eV\cdot m)^2}{2(5E5\, eV)}\left(\frac{2\pi}{10^{-2} m}\right)^2$

$\hookrightarrow 6.28 \times 10^2\, m^{-1}$

$\hookrightarrow = 1.6E{-14}\, eV$

$\hbar c = 2 \times 10^{-7}\, eV \cdot m$
$mc^2 = 5 \cdot 10^5\, eV$ (for e^-)
$n = 4.22 \times 10^{27}\, m^{-3}$

c) If a band has width 0.2 eV, is the approximation of treating the electron states in the band as a continuum justified, given this energy difference? What would be your argument?

$0.2\, eV \gg 1.6E{-14}\, eV$

Since a bandwidth of 0.2eV would allow for an insanely large amount of states across the width, treating the width as a continuum is physically accurate.